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Obesity, Diabetes, and Physical Activity

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Dan Donlevie is a 10-time successful Ironman triathlete and a two-time ultramarathon finisher. He is also a type 1 diabetic. At age 40, Dan began experiencing a series of worsening symptoms indicative of diabetes—excessive thirst, frequent urination, and unexplained weight loss. At first, Dan chalked these symptoms up to the intense training regimen necessary for a competitive endurance athlete, but when his blood glucose concentration was found to be over 600 mg/dl (normal fasting values are 80 to 120 mg/dl), he was diagnosed with adult-onset type 1 diabetes. Overcoming the negative feelings associated with this surprise diagnosis, Dan did two amazing things. He trained, modified his diet, and managed his disease to the extent that he not only completed the U.S. Ironman Championship in New York City 5 months later, he set a personal record. Second, he became a certified diabetes lifestyle coach dedicated to helping other athletes take control of their diabetes while continuing their active, competitive lifestyles. Many notable athletes have type 1 diabetes, including Jay Cutler (American football), Chris Dudley (basketball), Gary Hall, Jr. (swimming), Wasim Akram (cricket), Missy Foy (marathon), and Ben Coker (soccer). Diabetes is indeed a disease; however, with the proper mindset, knowledge base, and approach to activity and diet, it can be successfully managed—even with the demanding life of an elite athlete.

While hundreds of millions of people around the world are malnourished, many Americans are dying as a direct or indirect result of *overconsumption* of food. Billions of dollars are spent each year on diets, gadgets, and other weight loss methods and additional billions for the health care costs associated with obesity. Another common, and sometimes related, chronic disease is type 2 diabetes mellitus, which links obesity and insulin resistance. Further, there are common connections between insulin resistance and coronary artery disease, hypertension, and many other diseases or disorders.⁹

A sedentary lifestyle has been associated with an increased risk for both obesity and diabetes, and both are strongly linked to other diseases such as coronary artery disease and many types of cancer. Furthermore, approximately one-third of all Americans are obese, have type 2 diabetes, or both. In this chapter, we focus on obesity and diabetes, discussing their prevalence, their etiology, the health problems associated with each disease, and general treatment

options. Finally, we consider the role that physical activity can play in their prevention and treatment.

Understanding Obesity

The terms *overweight* and *obesity* are often used interchangeably, but technically they have different meanings.

Terminology and Classification

Overweight and obesity are both conditions related to the accumulation of excess adipose (fat) tissue. In the past, these conditions were defined using standard height/weight tables to establish norms. These tables were established and published by insurance companies in the 1950s and based on actuarial rather than physiological data, but they are used less frequently today. Other, more objective measures of body composition—such as body mass index, percent fat, lean body mass, and hip-to-waist ratio—are now more commonly used. Perhaps the most commonly used measure of overweight and obesity is body mass index.

In the older standard tables, normative values for weight were based on population averages. For this reason, a person could be overweight according to these standards and yet have a lower-than-normal body fat content. For example, football players may be overweight according to both the old weight tables and body mass index standards, yet many are typically much leaner than other people of the same age, height, and weight. Still other individuals are within the normal range of body mass index yet are overweight or obese.

Obesity refers to the condition of having an excessive percentage of body fat. This implies that the actual amount of body fat or its percentage of the total weight must be assessed or estimated (see [chapter 15](#) for assessment techniques). Exact standards for unhealthy fat percentages have not been established. However, men with more than 25% body fat and women with more than 35% are typically considered obese. Men with relative fat values of 20% to 25% and women with values of 30% to 35% are considered **overweight**. Allowances are higher for women because of sex-specific fat deposits such as breast tissue and hips, buttocks, and thighs, as will be discussed later in the chapter.

Despite its pitfalls, **body mass index (BMI)** is the most widely used clinical standard to estimate obesity because of its simplicity. To determine a person's BMI, body weight in kilograms is divided by the square of body height in meters. As an example, a man who weighs 104 kg (230 lb) and is 1.83 m (6 ft) tall would have a BMI of 31 kg/m² [104 kg/(1.83 m)² = 104 kg/3.35 m² = 31 kg/m²]. Generally, the BMI is highly correlated with body fat and usually provides a reasonable estimate of obesity. [Table 22.1](#) provides a simple way of determining BMI from height and weight.

The classification system based on BMI proposed by the World Health Organization was adopted by the National Institutes of Health in 1998 with several modifications and has been in use since 2000.²⁷ In [table 22.2](#), BMI values are divided into five categories: underweight, normal weight, and overweight, along with three classes of obesity—classes I, II, and III. *Morbid obesity* usually refers to those persons with BMI >40.0.¹⁸ The degree of disease risk in [table 22.2](#) is determined by both BMI and waist circumference. Waist circumference is important because it reflects visceral fat, which imparts higher risk for mortality, as discussed later in the chapter. A larger waist circumference increases risk within any given BMI category. Racial and ethnic differences affect the relation between BMI and fatness, necessitating different BMI cut-points for overweight and obesity in specific groups. As an example, a number of studies have indicated that for a given BMI, Asians have a higher percentage body fat. Accordingly, for the same BMI, the health risk is higher in Asian men and women. However, a BMI of 30 or higher almost always indicates excessive adiposity or obesity across all populations, regardless of race or ethnicity.

TABLE 22.1 **Body Mass Index**

	BMI																
BMI	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
Height (ft and in.)	Weight (lb)																
4'10" (58")	91	96	100	105	110	115	119	124	129	134	138	143	148	153	158	162	167
5' (60")	97	102	107	112	118	123	128	133	138	143	148	153	158	163	168	174	179
5'2" (62")	104	109	115	120	126	131	136	142	147	153	158	164	169	175	180	186	191
5'4" (64")	110	116	122	128	134	140	145	151	157	163	169	174	180	186	192	197	204
5'6" (66")	118	124	130	136	142	148	155	161	167	173	179	186	192	198	204	210	216
5'8" (68")	125	131	138	144	151	158	164	171	177	184	190	197	203	210	216	223	230
5'10" (70")	132	139	146	153	160	167	174	181	188	195	202	209	216	222	229	236	243
6' (72")	140	147	154	162	169	177	184	191	199	206	213	221	228	235	242	250	258
6'2" (74")	148	155	163	171	179	186	194	202	210	218	225	233	241	249	256	264	272
6'4" (76")	156	164	172	180	189	197	205	213	221	230	238	246	254	263	271	279	287

1 lb = 0.454 kg, 1 in. = 2.54 cm.

Reprinted from NIH/National Heart, Lung, and Blood Institute (NHLBI), Evidence Report of Clinical Guidelines on the Identification, Evaluation, and Treatment of Overweight and Obesity in Adults, 1998.

TABLE 22.2 **Classification of Overweight and Obesity by BMI, Waist Circumference, and Associated Disease Risk^a**

Classification	BMI (kg/m ²)	Obesity class	Disease risk ^b	
			Men ≤37 in. (94 cm) Women ≤32 in. (80 cm)	Men >40 in. (102 cm) Women >35 in. (88 cm)
Underweight	<18.5			
Normal ^c	18.5-24.9			
Overweight	25.0-29.9		Increased	Substantially increased
Obesity	30.0-34.9	I	High	Very high
	35.0-39.9	II	Very high	Very high
Extreme obesity	≥40	III	Extremely high	Extremely high

^aDisease risk for type 2 diabetes, hypertension, and cardiovascular disease.

^bRelative to normal weight and waist circumference.

^cIncreased waist circumference also can be a marker for increased risk even in persons of normal weight.

Adapted from World Health Organization (2016).

Historically, the percentage of adults who were overweight or obese varied tremendously, depending on what cut-points or standards were used to classify people as overweight and obese. We now better understand the true prevalence of overweight and obesity and how it has changed over time. Further, including the overweight category has proven useful, providing a gray area between normal weight and obesity. People in this category can either have an above-average fat-free body mass, such as the football players mentioned previously, or can be marginally overfat. However, as stated earlier, beyond a BMI of 30, individuals are likely to be obese.

Prevalence of Overweight and Obesity

The prevalence of overweight and obesity in the United States has increased dramatically since the 1960s. According to the Centers for Disease Control and Prevention,²⁸ the overall prevalence of obesity in the United States was just over 30% for the period from 2011 to 2014. As illustrated in [figure 22.1](#), the prevalence of obesity is lower in 20- to 39-year-olds but increases beyond age 40 for both men and women.

If we include the category of overweight (those with BMI values of 25 and above consistent with the World Health Organization and National Institutes of Health classification systems), more than 70% of men and over 64% of women in the U.S. adult population are overweight or obese. The prevalence of overweight and obesity has increased by 68% and 142%, respectively, since 1976, and the prevalence of morbid obesity (class III) has increased by 360%.²³

The most recent published data from the U.S. population show that there were no significant overall changes in either adults or youth with respect to overweight and obesity over the decade from 2003-2004 to 2011-2012. However, the prevalence of obesity across the U.S. population remains very high.

When we look at these data by race, it is apparent that the problem is much more significant in Hispanic men and women and in non-Hispanic black women ([figure 22.2](#)).

These trends are not unique to the United States. Canada, Australia, and most of Europe have seen similar increases in obesity prevalence.³⁹ The most recent data show that obesity is spreading to most regions of the world. [Table 22.3](#) provides estimates of the rates of obesity (combined for men and women) for select countries around the world. These data may be somewhat misleading since there is a great variation in the dates of these surveys; however, indications are that rates of obesity have continued to skyrocket during the past 10 years, resulting in a worldwide obesity epidemic. Most recent estimates project that approximately 2 billion people worldwide are overweight or obese.

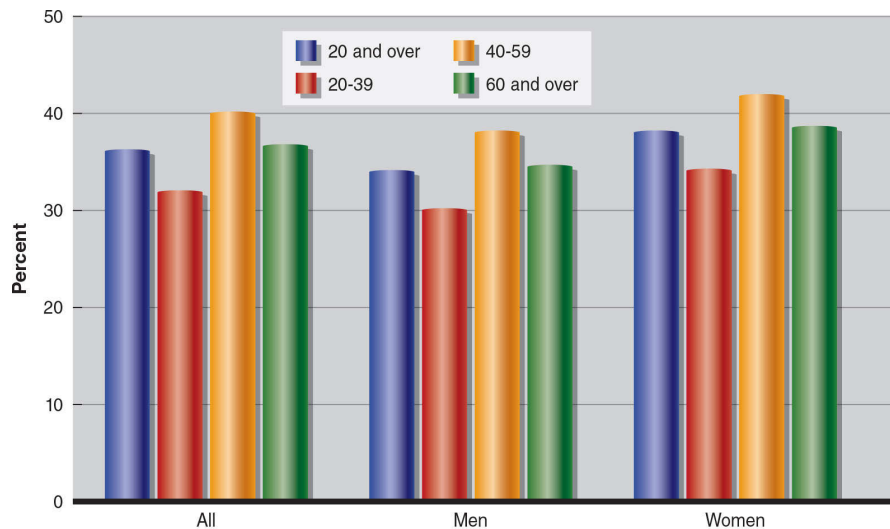


FIGURE 22.1 Prevalence of obesity in the United States among adults aged 20 and older, 2011 to 2014.

Data from Center for Disease Control and Prevention (2011).

Unfortunately, this same trend of increasing prevalence of overweight and obesity has been reported in U.S. children and adolescents. [Figure 22.3](#) illustrates the trends in the prevalence of overweight from 1971 through 2012 for preadolescent and adolescent boys and girls. Because BMI is much less precise with respect to estimating body fat in children and adolescents, scientists typically use the cut-point for BMI of greater than the 95th percentile, a value that likely indicates that the child is overfat. The prevalence of overweight remained relatively constant from 1971 through 1980, increased dramatically from 1980 through 2004, and now seems to be leveling off. The other good news, perhaps, is that the prevalence of obesity in 2- to 5-yearolds significantly decreased from 14% in 2003-2004 to 8.4% in 2011-2012.

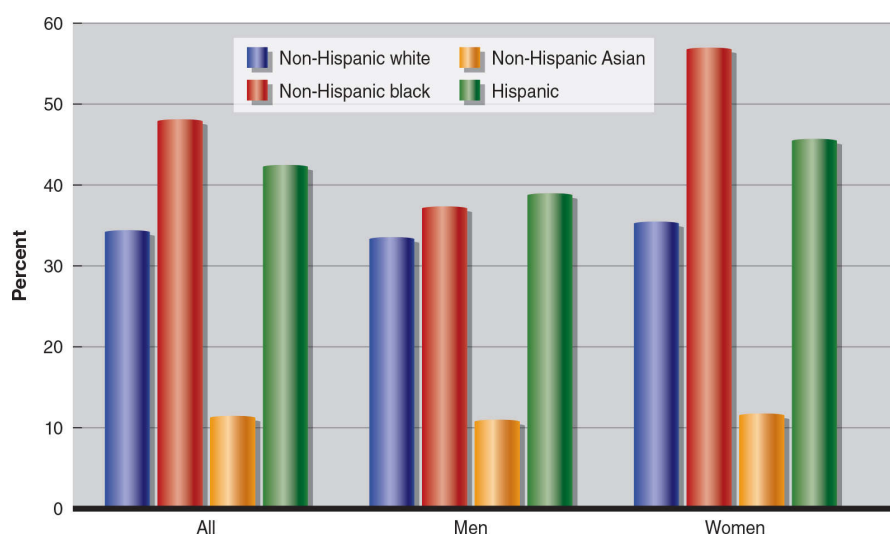


FIGURE 22.2 The prevalence of obesity in U.S. adults aged 20 and older by sex, race, and Hispanic origin, 2011 to 2014.

Data from Ogden et al. (2015).

TABLE 22.3 Prevalence of Adult Obesity in Selected Countries

Country	Obesity prevalence
North Korea	2.4%
Japan	3.3%
India	4.9%
South Korea	5.8%
Switzerland	19.4%
Netherlands	19.8%
Brazil	20.0%
Germany	20.1%
Italy	21.0%
Spain	23.7%
France	23.9%
Russian Federation	24.1%
Israel	25.3%
Canada	28.0%
Mexico	28.1%
United Kingdom	28.1%
Australia	28.6%
Turkey	29.5%
Saudi Arabia	34.7%
Kuwait	39.7%
Cook Islands	50.8%

Data from OECD (2014).

The average person in the United States gains approximately 0.3 to 0.5 kg (0.7 to 1.1 lb) of weight each year after age 25. This seemingly small gain results in an average of 9 to 15 kg (15 to 33 lb) of excess weight by age 55. At the same time, bone and muscle mass decrease by approximately 0.1 kg (0.22 lb) per year due to the combined effects of reduced physical activity and normal aging processes. Taking this into account, an average person's fat mass

actually increases by about 0.4 kg (0.9 lb) each year, which equates to a 12 kg (27 lb) fat gain over a 30-year period! (It should be noted that these values are approximations and that they vary by sex and race or ethnicity.)

However, population estimates of increases in calorie intake and the decreases in activity that have accompanied modern life do not match this average weight gain over a 30-year period.¹⁶ The estimated increase in caloric intake coupled with decreases in physical activity should have accounted for annual weight gains of 39.1 kg (85.8 lb) and 37.2 kg (82 lb) for men and women, respectively. Thus, the adages often heard in weight management programs that “a calorie is a calorie” and 3,500 kcal = 1 lb does not accurately reflect real-life weight gains or the weight loss experienced by those in (or out of) weight management programs. Other factors must be at work that modify the energy balance equation.¹⁵

A major concern in light of the increasing rates of obesity, coupled with the earlier onset of obesity, is the impact that this will have on individual and national health care. With the onset of obesity occurring at younger ages will come an increased accumulative exposure to excess weight, and this will likely precipitate an earlier onset of obesity-related diseases such as diabetes.

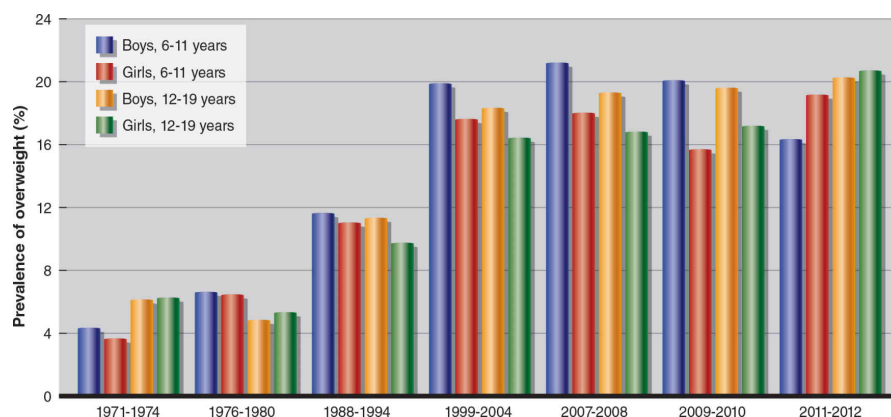


FIGURE 22.3 The increasing prevalence of overweight (>95th percentile body mass index [BMI] for age) in children and adolescents in the United States from 1971 through 2012.

Data from Flegal et al. (2002).

Control of Body Weight

One must know how body weight is controlled or regulated to better understand how a person becomes obese. Body weight regulation

has puzzled scientists for years. The average male consumes about 2,500 kcal per day, or nearly 1 million kcal per year. An average gain of 0.4 kg (0.9 lb) of fat each year represents an imbalance of only 3,111 kcal per year between energy intake and expenditure (3,500 kcal is the energy equivalent of 0.45 kg, or 1 lb, of adipose tissue). This translates to a surplus of less than 9 kcal per day. Even with a weight gain of 1.5 lb (0.7 kg) of fat per year, the body can balance caloric intake to within one potato chip per day of what is expended, a truly remarkable example of homeostasis.

In the past, the body's ability to balance its caloric intake and expenditure to within such a narrow range led scientists to believe that body weight is regulated around a given set point for body weight, similar to the way core temperature is regulated. More recently, research has centered on the regulation of caloric intake and expenditure, collectively called "energy balance." The physiological drive to control and regulate energy balance, as well as to conserve fuel in the form of stored fat, may be at the core of the obesity problem.^{16,26} People become overweight when they have a positive energy balance. That is, when caloric intake exceeds caloric expenditure over an extended period of time, excess adipose tissue is accumulated.

Weight management programs center on losing weight (and fat) by establishing a negative energy balance in which caloric expenditure exceeds caloric intake over an extended period of time. It is critical to understand, however, that a physiological control system (and thus the control of energy balance) is affected by changes in, and regulation of, each of these components—energy intake and energy expenditure. So changes in dietary pattern and nutrient consumption, in physical activity, and ultimately in weight, will change the physiology of energy balance and probably the control system itself. Thus, sustained weight loss can be exceedingly difficult because of the physiological adaptations that occur with long-term negative caloric balance and physiological change that occur during any attempt to modify calorie expenditure and intake.

Energy balance is established and controlled within the three components of energy expenditure (see [figure 22.4](#)):

1. Resting metabolic rate (RMR)

2. The thermic effect of a meal (TEM)
3. The thermic effect of (physical) activity (TEA)

Recall that RMR, as discussed in [chapter 5](#), is the body's metabolic rate early in the morning following an overnight fast and 8 h of sleep. The term *basal metabolic rate (BMR)* also is used but generally implies that the person fasted for 12 to 18 h and that the BMR measurement was made immediately upon waking. Most research today reports the RMR represents the minimal amount of energy expenditure needed to support basic physiological processes. It accounts for about 60% to 75% of the total energy we expend each day.

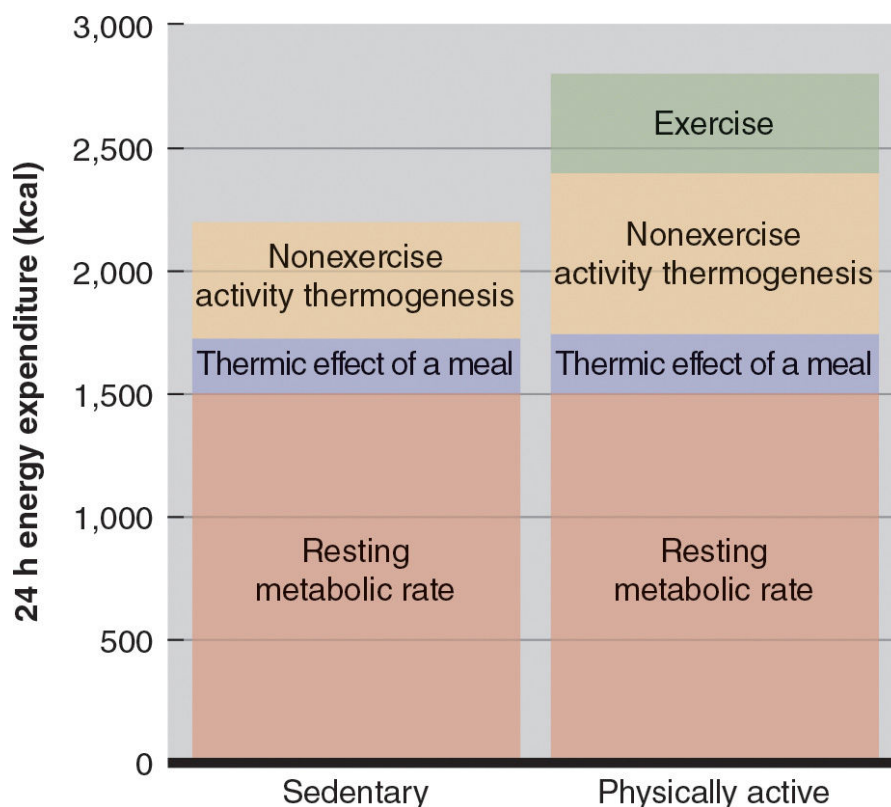


FIGURE 22.4 The components of energy expenditure, showing examples for a sedentary person and a physically active person. See the text for a detailed explanation.



The **thermic effect of a meal (TEM)** represents the increase in energy expenditure that is associated with the digestion, absorption, transport, metabolism, and storage of ingested food. While TEM peaks during and after food ingestion, on average the TEM accounts for approximately 8% to 10% of our total energy expenditure each day. This value also includes some energy waste, because the body can increase its metabolic rate above that necessary for food processing and storage. We seldom notice the TEM; however, after a very large holiday meal with family, people often start feeling warm and drowsy, perhaps indicating that the metabolic rate has increased perceptively. The TEM component of metabolism might be defective in some people with obesity, possibly attributable to a defect in the energy-wastage component, leading to a surplus of calories.

The **thermic effect of activity (TEA)** is simply the energy expended above the RMR to accomplish a given task or activity, whether it is sitting and typing on a computer or running a 10 km race. The TEA accounts for the remaining 15% to 30% of our energy expenditure. This is the most variable component of energy expenditure and the component that can most easily be purposefully modified.

The body adapts to major increases or decreases in energy intake by altering the energy expended by each of these three components—RMR, TEM, and TEA. With fasting or very low-calorie diets, all three components of total energy expenditure decrease in an attempt to conserve energy stores. This is dramatically illustrated by reported decreases in RMR of 20% to 30% or more within several weeks after initiating a fast or consuming a very low-calorie diet (<800 kcal/day). Conversely, all three components of energy expenditure increase somewhat with overeating. In this case, the body appears to be trying to prevent unnecessary storage of the surplus calories. These adaptations appear to be controlled by changes in the sympathetic nervous system, the endocrine system, and other physiological and biochemical systems that control both metabolism and appetite. The systems are extremely complex and difficult to study because of the large number of interacting variables and cross talk between systems.¹⁵ This remains a critical area for ongoing and future research.

The composition of the diet (nutrients) and physical activity (including exercise and nonexercise physical activity) are primary control points for the energy balance equation. It is clear that changes in fat and carbohydrate (especially intake of simple or refined carbohydrates) content of the diet, coupled with a decrease in physical activity levels over an extended period of time, change our physiological processes markedly.

In Review

- Overweight is a body weight that exceeds the standard weight for a certain height and frame size. Obesity refers to excessive body fat, meaning more than 25% body fat for men and more than 35% body fat for women.
- To calculate a person's BMI, we divide body weight in kilograms by the square of height in meters. This value is highly correlated with relative body fat and provides a reasonable estimate of obesity. Body mass index values of 25.0 to 25.9 correspond to overweight, and values of 30.0 or higher correspond to obesity.
- Prevalence of obesity and overweight in the United States has increased dramatically since the 1970s.
- The average person gains 0.3 to 0.5 kg (0.7 to 1.1 lb) per year after age 25 but also loses 0.1 kg (0.22 lb) of fat-free mass per year, meaning a net gain of 0.4 kg (0.9 lb) of fat each year.
- Our bodies attempt to regulate body weight through a regulatory system based on energy balance and total energy expenditure.
- Daily energy expenditure is reflected by the sum of the RMR, the TEM, and the TEA. The body adapts to changes in energy intake by adjusting all of these components.

Etiology of Obesity

At various times throughout human history, obesity was thought to be caused by basic hormonal imbalances resulting from failure of one or more of the endocrine glands to properly regulate body weight. At other times, it has been believed that gluttony, rather than glandular malfunction, was the primary cause of obesity. In the first case, a person is perceived as having no control over the situation, yet in the second, he or she is held directly responsible. Findings from recent research show that obesity most likely results from a combination of many factors. Thus, the etiology, or cause, of obesity is extremely

complex and multifactorial, rather than a simple matter of genes, hormones, or even personal choice.

Experimental studies on animals have linked obesity to heredity. Studies of humans have also shown a direct genetic influence on height, weight, and BMI. Studies by Dr. Claude Bouchard have provided possibly the strongest evidence yet of a significant genetic component for obesity.^{4,5} He and his coinvestigators took 12 pairs of young adult male monozygotic (identical) twins and housed them in a closed section of a dormitory under 24 h observation for 120 consecutive days. The subjects' diets were monitored during the initial 14 days to determine their baseline caloric intake. Over the next 100 days, the subjects were fed 1,000 kcal above their baseline consumption for 6 of every 7 days. On the seventh day, the subjects were fed only their baseline diet. Thus, they were overfed by 1,000 kcal per day for 84 of the 100 days. Activity levels were also tightly controlled. At the end of the study period, the individual weight gained varied widely, from 4.3 to 13.3 kg (9.5-29.3 lb)—a threefold variation in weight gain for overconsumption of the same number of calories. However, the response of both twins in any twin pair was quite similar. The same association was found for gains in fat mass, percentage body fat, and subcutaneous fat. These and subsequent studies have demonstrated that genetics plays a major role in determining one's *susceptibility to becoming* obese. But the data demonstrate that other factors are also responsible since every subject in this study gained at least 4 kg.

Hormonal imbalances, emotional issues, and alterations in basic homeostatic mechanisms all have been shown to be either directly or indirectly related to the onset of obesity. Environmental factors, such as cultural habits, food availability, inadequate physical activity, and improper diets, are also major factors related to obesity.

While research confirms that there is a significant genetic component in the etiology of obesity, it is clearly possible to be obese based simply on lifestyle, with or without a family history (genetics) of obesity. It is also possible to be relatively lean, even with a genetic predisposition to obesity, through a healthy combination of diet and physical activity.³³

Thus, the origin and etiology of obesity are highly complex, and the specific causes undoubtedly differ from one person to the next. Recognizing this is important for treating existing obesity and for preventing its onset. To attribute obesity solely to a single cause such as overeating or inactivity underestimates the cause and the magnitude of the problem.

Energy Balance and Obesity

In a 2012 review article, Dr. James Hill and colleagues¹⁶ discuss the physiology (and neurophysiology) of weight gain, weight loss, and obesity. Energy balance, the sum of calorie expenditure and calorie intake, is the key to weight loss or the prevention of weight gain. Establishing a negative energy balance (caloric expenditure > caloric intake) is required for weight loss to occur. Our ancestors had a very high daily level of physical activity, coupled with a high drive to consume calories in order to establish that balance and to maintain a stable weight. Thus, humans seem to be wired to maintain weight (and energy balance) best with high levels of energy expenditure, not with large decreases in energy intake. Weight gain underlying the current obesity epidemic has been postulated as our body's attempt to reestablish energy balance in the face of excess food intake. With an increased body weight comes an increased metabolic rate and an increased caloric cost of most activities; however, with large weight gains and low activity levels, we cannot achieve energy balance, and the result is excess adipose tissue deposition. Thus, the adaptation that is most easily achieved is accumulating excess adipose tissue (thus added body weight) and becoming overweight and eventually obese. The fact is that becoming obese increases energy expenditure and eventually reestablishes the energy balance.¹⁶

Increases in caloric consumption with concurrent decreases in physical activity (both occupational and recreational) over the past 50 to 100 years have resulted in a high prevalence of overweight and obesity in the U.S. population. In 1960, the prevalence of overweight and obesity in the U.S. population was approximately 45%. By 2008, the prevalence was 68%. This large increase was due to a positive energy balance in a large portion of the population.¹⁶

Obesity is not simply a result of a problem with one of the components of the energy balance equation. Since 1960, the average

individual food consumption has increased significantly, by 168 kcal/day for men and 335 kcal/day for women. There have been concomitant decreases in the average caloric expenditure of occupational activities, as well as decreased voluntary daily physical activity among men and women. When this decrease is combined with the average increase in calorie consumption, the positive energy balance would predict an average annual weight gain over the past 50 years that is much greater than that actually recorded. The actual average total weight gain for men and women has been about 30 and 25 lb (13.6 and 11.3 kg), respectively, not the amounts noted that would have been predicted by estimates of population-based changes in energy intake and expenditure. Thus, there must be additional physiological factors at work in the obesity epidemic. These are the adaptations that occur in the extremely complex physiology that controls energy balance and therefore body weight.

Jean Mayer was a preeminent Harvard physiologist who studied the physiology of nutrition and hunger. Mayer hypothesized that energy intake (caloric intake) is matched to calorie expenditure but only at high levels of energy expenditure. He further hypothesized that regulation of calorie intake is much less sensitive to energy expenditure at low levels of energy expenditure; thus, chronically maintaining a positive energy balance (calorie intake > calorie expenditure) is physiologically very difficult without weight gain. Therefore, the control of energy balance in humans works well at high levels of physical activity, but not at lower levels. So, in active and very active people, weight gain is often not a problem. But in sedentary individuals, the difficulty in sufficiently decreasing caloric intake to match very low levels of physical activity ultimately results in weight gain and obesity. It also appears that low metabolism (presumably meaning low RMR) does not significantly contribute to weight gain and obesity. Though this concept is not entirely supported by the literature, a number of well-accepted studies do support the plausibility of this view.

The most efficient way to prevent weight gain (and probably to lose weight) is to mimic those preindustrial high levels of energy expenditure in order to balance the higher average caloric intake. And most people simply cannot balance today's historically low levels of

physical activity by consuming fewer calories. As a homeostatic mechanism, excess adipose tissue elevates RMR and increases the caloric cost of physical activity. Even with this physiological adaptation (increased stores of energy in the form of fat), energy balance remains positive and weight is gained as increased caloric expenditure simply cannot keep pace with the increased food intake.

Health Problems Associated with Overweight and Obesity

Overweight and obesity are associated with an increased overall mortality.⁶ This relationship is curvilinear, as shown in [figure 22.5](#). A major increase in the risk of death occurs when the BMI exceeds 30 kg/m², although BMI values between 25.0 and 29.9 are associated with an increased morbidity risk for many diseases. A number of more recent studies have reported that excess mortality is primarily associated with BMI values of 35.0 and above.

Excess morbidity and mortality associated with obesity and overweight are linked with a number of major diseases, including the following:⁷

- Hypertension
- Dyslipidemia
- Type 2 diabetes
- Coronary artery disease
- Stroke
- Gallbladder disease
- Osteoarthritis
- Sleep apnea and breathing problems
- Some cancers (endometrial, breast, colon, kidney, gallbladder, and liver)
- Mental illness such as clinical depression, anxiety, and other mental disorders

With the large increase in the prevalence of obesity in the United States since the 1970s, it is not surprising to also see a very high prevalence of metabolic syndrome in U.S. adults. **Metabolic syndrome** is a cluster of related conditions—increased blood pressure, elevated blood glucose, excess abdominal body fat, and elevated cholesterol or triglycerides—that occur together, increasing

the risk of heart disease, stroke, and diabetes. Metabolic syndrome is defined as (1) waist circumference ≥ 102 cm (male adults) and ≥ 88 cm (female adults), (2) fasting plasma glucose ≥ 100 mg/dl, (3) blood pressure of $\geq 130/85$ mmHg, (4) triglycerides ≥ 150 mg/dl, and (5) high-density lipoprotein cholesterol (HDL-C) < 40 mg/dl (male adults) and < 50 mg/dl (female adults). One-third of all adults in the United States meet these criteria. Data from the National Health and Nutrition Examination Survey (1988-2012) revealed that the prevalence of metabolic syndrome rose from 25% in 1988-1994 to 34% in 2007-2012.²⁵ Non-Hispanic black men were less likely than non-Hispanic white men to have metabolic syndrome, but the exact opposite relation is true for women.

Changes in Normal Body Function

The prevalence and extent of changes in physiological function associated with excess adiposity vary with the individual and with the degree of fatness. Conditions and diseases commonly associated with obesity (especially morbid obesity) include respiratory problems (sleep apnea, chronic pulmonary disease), abnormal blood clotting (thrombosis), enlargement of the heart, and congestive heart failure. Obese individuals (and often those who are overweight) typically have decreased exercise tolerance and a lower level of physical fitness due to decreased levels of physical activity. Additional weight gain (and excess adiposity) further reduces activity levels and exercise tolerance decreases. This is a vicious, positive feedback cycle of abnormal function associated with excess adiposity and decreased levels of physical activity and physical fitness, which leads to more inactivity and more pathophysiology. This cycle affects millions of people in the United States and around the world.

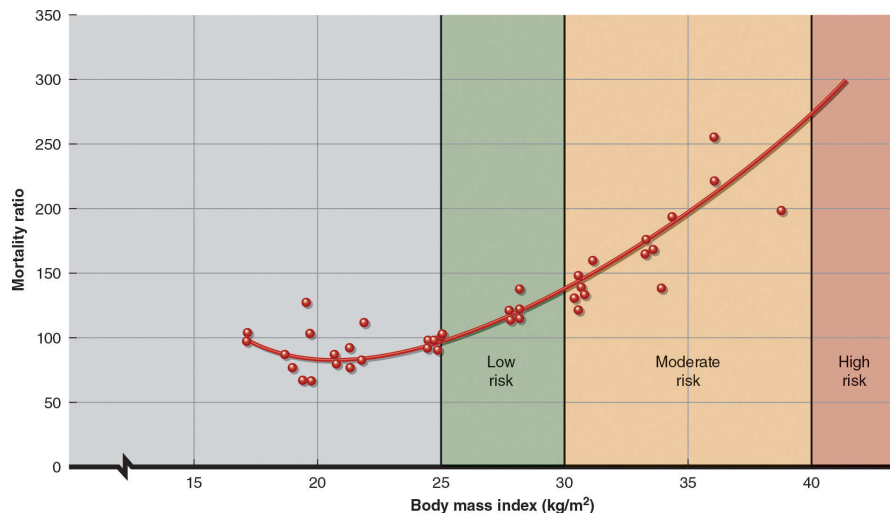


FIGURE 22.5 The relation between body mass index and excess mortality. A mortality ratio of 100 represents average mortality. The lowest portion of the curve (body mass indexes under 25) indicates very low risk, while mortality risk increases disproportionately with increasing BMI.

Bray GA. Obesity: Definition, diagnosis and disadvantages. *Med J Aust* 1985; 142: S2-S8. © Copyright 1985 The Medical Journal of Australia - reproduced with permission. The Medical Journal of Australia does not accept responsibility for any errors in translation.

Increased Risk for Certain Diseases

An increased risk of developing certain chronic diseases also is associated with obesity. Both hypertension and atherosclerosis are directly linked to obesity (see [chapter 21](#)), as are impaired carbohydrate metabolism and type 2 diabetes. Obesity and type 2 diabetes are particularly common comorbid diseases.

Since the 1940s, major sex differences in the way in which fat is stored or patterned on the body have been recognized. As shown in [figure 22.6](#), men tend to store fat in the upper body, particularly the abdominal area, whereas women tend to store fat in the lower body, particularly the hips, buttocks, and thighs. Abdominal obesity is closely associated with visceral obesity. Obesity that follows the male pattern is referred to as **upper body (android) obesity**, or apple-shaped obesity, and the female pattern is referred to as **lower body (gynoid) obesity**, or pear-shaped obesity. Apple-shaped obesity is more closely attributable to visceral fat and thus carries a higher risk.

Research beginning in the late 1970s and early 1980s established upper body obesity as a risk factor for the following conditions:

- Coronary artery disease
- Hypertension

- Stroke
- Elevated blood lipids
- Diabetes

Furthermore, abdominal obesity appears to be more important than total body fatness as a risk factor for these diseases. Waist and hip circumference, or girth measurements, can be used to identify people with increased risk. A waist/hip girth ratio greater than 0.90 for men and greater than 0.85 for women indicates increased risk. With upper body obesity, the increased risk may result from the close proximity of visceral fat depots to the portal circulatory system (circulation to the liver). [Figure 22.7](#) shows a young woman being placed into a computed tomography (CT) scanner in order to assess visceral abdominal fat ([figure 22.7a](#)) and MRI scans of the upper thigh of two men ([figure 22.7b](#)).³² The subject in the lower two scans of [figure 22.7b](#) illustrates the effect of muscular atrophy and the comparison of visceral (deep) fat versus subcutaneous fat.



Upper body (android) obesity



Lower body (gynoid) obesity

FIGURE 22.6 Obesity patterns tend to differ by sex.

Effects of Obesity on Established Diseases

Though the effects of obesity on existing diseases are not totally clear, obesity clearly contributes to progression and exacerbation of some medical conditions, including type 2 diabetes, hypertension, and heart disease. Weight loss is typically prescribed as an integral part of the treatment plan. Conditions that generally benefit from weight reduction include

- angina pectoris,
- hypertension,
- congestive heart disease,

- myocardial infarction (reduced risk of recurrence),
- varicose veins,
- type 2 diabetes, and
- orthopedic problems.



FIGURE 22.7 (a) A subject in the process of undergoing a computed tomography (CT) scan. (b) Colorized MRI scans of the upper thigh in two people. The subject in *the bottom scans* has considerably more fat infiltration (light areas).

Weight Loss

Healthy weight loss should generally not exceed 0.45 to 0.9 kg (1-2 lb) per week, and greater rates of weight loss should be managed only with medical supervision. Losing just 0.45 kg (1 lb) per week will result in the loss of 23.4 kg (52 lb) in only 1 year! Few people become obese that rapidly, so weight loss should be considered a long-term undertaking. Rapid weight loss is almost always short lived, and the weight is regained quickly. Rapid weight loss involves large losses of body water, and because the body has built-in mechanisms to prevent an imbalance in body fluid levels, the lost water eventually will be replaced. Thus, a person wishing to lose 9 kg (20 lb) is advised to attempt to attain this goal over a period of 3 to 5 months.

RESEARCH PERSPECTIVE 22.1

The Fat-but-Fit Paradox

In the past 30 years, a multitude of longitudinal studies have convincingly demonstrated that low cardiorespiratory fitness (below first quintile = 20th percentile) increases all-cause mortality as well as cardiovascular disease

mortality. Obesity is also a primary risk factor for all-cause and cardiovascular disease mortality. Interestingly, it has been suggested that being fit may alleviate some of the negative consequences of obesity, a concept known as the fat-but-fit paradox. That is, the mortality risk in obese individuals who are fit is not different from the risk in normal-weight and fit adults (theoretically, the healthiest group possible). In addition, several studies have now suggested that normal-weight but unfit individuals could be at a higher risk than obese but fit individuals.

Though this emerging evidence indicates that moderate-to-high cardiorespiratory fitness may mitigate some of the detrimental health consequences of obesity, many research questions remain.³⁰ For example, the available evidence supports a direct link between cardiorespiratory fitness and cardiometabolic risk factors in children and adolescents. Limited reports indicate that higher levels of fitness may attenuate the consequences of excess adiposity in children; however, these types of studies must be expanded to include more participants. This is especially important given the increasing public health concerns related to childhood obesity.

Perhaps the most important studies that remain to be conducted regarding the fat-but-fit paradox are randomized controlled trials. To date, all available evidence for the fat-but-fit paradox comes from observational studies. Although the information derived from these studies is valuable, it cannot determine cause and effect. Thus, randomized controlled trials are necessary to determine whether exercise interventions in obese individuals reduce cardiovascular disease risk even in the absence of weight loss. Such studies will further our understanding of the fat-but-fit paradox, because they will allow researchers to determine whether improvements in health without weight loss are driven by improvements in cardiorespiratory fitness. Public health strategies targeting obese adults should target both weight and fat reduction as well as increases in cardiorespiratory fitness, especially for unfit individuals.

VIDEO 22.1 Presents Louise Burke on successful weight loss approaches and why some people have difficulty losing weight.



Many special diets have achieved popularity over the years, with each claiming to be the ultimate in effective and manageable weight loss. Some diets have been developed for use either in the hospital or at home under the supervision of a physician. These are often referred to as *very low calorie diets*, since they allow less than 800 kcal of food per day. Most of these have been formulated with proportions of protein and carbohydrate aimed at minimizing the loss of fat-free body mass. Many of these are effective, but no single diet has been shown to be more effective than any other. Again, the crucial factor for all diets should be the development of a caloric deficit while maintaining a complete, balanced diet that meets the body's nutrition needs. The best diet is the one that meets these simple criteria and is best suited to the individual.

Because improper eating habits are at least partially responsible for most cases of obesity, no diet should be viewed as a stand-alone solution or quick fix. A person should learn to make permanent changes in dietary habits, especially reducing the intake of fat and simple sugars. For most people, simply eating a low-fat diet will gradually reduce weight to a desirable level. One potential problem with low-fat diets, however, is that people mistakenly assume that low fat equates to low calorie, and this often is not the case. For most people, simply reducing total caloric intake by 250 to 500 kcal per day, combined with selecting low-fat foods and foods low in simple sugar, would be sufficient to accomplish their desired weight loss goals.

Pharmaceutical approaches to obesity have also been used to assist patients in weight loss by decreasing their appetite or increasing their RMR. Unfortunately, a number of side effects are associated with these drugs, some of which are very serious and can be life threatening. Surgical techniques also are used to treat extreme obesity but only as a last resort when other treatment procedures have failed and the obesity is life threatening. Gastric bypass surgery and gastric banding are the most common surgical procedures used to treat morbid obesity. Both restrict the amount of food that can enter the stomach. While highly effective, these procedures are expensive and have associated risks, although the average mortality rate is about 1% to 2%. These procedures have also been shown to

moderate chronic diseases that are associated with obesity, including type 2 diabetes. Generally, surgery should be reserved for people who have morbid obesity or obesity with significant associated risk factors.

Behavior modification is one of the most effective techniques for helping people with weight problems. Major weight losses have been achieved through changes in basic behavior patterns associated with eating. Furthermore, these weight losses appear to be more permanent than those associated with other weight loss strategies. This approach appeals to most people because the techniques seem to make sense and are often easy to incorporate into a normal daily routine. For example, an individual might not have to consciously reduce the amount of food eaten but simply agree that all eating will be done in one location, which often cuts down on snacking. Or an individual might be allowed to take as much food as desired with the first helping but have no second helpings. Many such simple changes can help regulate eating behavior and result in substantial weight loss.

Management Guidelines for Overweight and Obesity

In 2014, new guidelines for the management of overweight and obesity were jointly published by the American Heart Association (AHA), the American College of Cardiology (ACC), and The Obesity Society.¹⁸ These new guidelines, published in concert with three other sets of guidelines by the AHA and the ACC (assessment of cardiovascular risk, lifestyle modification to reduce cardiovascular risk, and the treatment of elevated cholesterol), were collectively designed to “promote optimal patient care and cardiovascular health.” As with most recent guidelines published by highly respected professional organizations, these guidelines are evidence based, focusing on critical questions that can be answered with published, evidence-based studies in the literature.

In this new set of guidelines, the critical questions centered on (1) the criteria that guide appropriate weight loss programs; (2) the health benefits and risks of weight loss with respect to amount of loss and duration of a program; (3) which strategies for dietary interventions

are effective; (4) the effectiveness of a comprehensive approach involving diet, exercise, and behavioral therapy; and (5) the safety and effectiveness of bariatric surgery as a primary therapeutic procedure. These documents are aimed at health care providers, primarily physicians and other gatekeepers, including advanced nurse practitioners and physician's assistants, but all allied health care professionals should be aware of such guidelines and employ them when appropriate.

Recommendations include the following:

- Assess height, weight, and waist circumference at least annually or more frequently if appropriate; determine BMI and advise therapeutic modality based on coronary heart disease risk.
- Counsel overweight and obese patients about cardiovascular risk factors and lifestyle changes that result in a sustained weight loss of 3% to 5% of body weight or more if possible. Greater amounts of weight loss result in greater benefits, but even modest amounts will reduce risk.
- Recommend a dietary pattern that restricts calories as part of a comprehensive lifestyle intervention. Prescribe 1,200 to 1,500 calories/day for women and 1,500 to 1,800 calories/day for men. There are effective, evidence-based diets that have been shown to promote healthy weight loss, but any intervention must reduce daily caloric intake. Among the effective diets are vegetarian and vegan, low fat (20% of calories from fat), low glycemic load, Mediterranean style, and low carbohydrate (<20 g/day without formal caloric restriction but with some total negative energy balance). Note that the common factor among these potentially effective diets is a negative energy balance.
- Because of individual variability, the strength of evidence (an important factor in determining both real-life efficacy and appropriateness of a diet for particular individuals) varies from high to low among these diets and other specific diet recommendations in these guidelines.
- A comprehensive program should include a prescribed, moderately reduced-calorie diet, increased physical activity,

and behavioral strategies that facilitate compliance with the changes.

- Bariatric surgery referral is recommended for adults with a BMI >40 or in adults with a BMI >35 who have comorbid (accompanying) conditions—for example, diabetes, hypertension, or dyslipidemia.

The published guidelines are extremely comprehensive, with detailed discussions about each of these recommendations, as well as extensive references for each of the statements. Exercise professionals should be familiar with these recommendations and guidelines for populations that they work with on a regular basis.

In Review

- The etiology of obesity is not simple; the cause can be any one or a combination of many factors.
- Studies of animals and humans indicate that there is a genetic component to obesity. The disorder also has been linked to hormonal imbalances, emotional trauma, homeostatic imbalances, cultural influences, physical inactivity, and improper diets. It is unlikely that a single cause is responsible.
- Overweight and obesity are associated with increased risk of general excess mortality.
- Obesity increases the risk of certain chronic degenerative diseases. Abdominal (visceral) obesity increases the risk of developing coronary artery disease, hypertension, stroke, elevated blood lipids, diabetes, and the metabolic syndrome. Also, obesity can worsen preexisting health conditions and diseases.
- Emotional or psychological problems may contribute to obesity, and the disorder itself, with the stigma it carries, can be psychologically damaging.
- In the treatment of obesity, it is important to remember that people respond differently to a given intervention. Some have large losses of weight in a relatively short period of time, while others appear to be resistant to weight loss and see only small losses.
- Weight loss generally should not exceed 0.45 to 0.9 kg (1-2 lb) per week. Simple diet modification, reducing intake of fat and simple sugars, is sufficient to help most people lose weight. Behavior modification is also an effective method of weight loss.

- The use of drugs or surgery in the treatment of obesity is generally not recommended unless deemed necessary for the patient's health by a physician.
-

Role of Physical Activity in Weight Management and Risk Reduction

Inactivity is a major cause of obesity in the United States. In fact, a sedentary lifestyle may be just as important in the development of obesity as overeating. Thus, increasing physical activity levels must be recognized as an essential component in any program of weight reduction or control.

RESEARCH PERSPECTIVE 22.2

Sedentary Behavior, Physical Activity, and Adiposity

Obesity is a traditional risk factor for cardiovascular disease. The type of fat is meaningful: excess visceral adipose tissue (VAT), intermuscular adipose tissue (IMAT), and liver fat increase cardiovascular risk, whereas subcutaneous adipose tissue (SAT) may be relatively less important in the pathophysiology of disease development.

A lack of regular physical activity is a primary factor contributing to obesity, and a sedentary lifestyle also contributes to greater cardiovascular risk. In fact, some evidence suggests that meeting the guidelines for physical activity may not be sufficient for disease prevention if accompanied by high levels of sedentary time. Sedentary behavior likely influences fat deposition, which would, of course, potentially explain part of the link between sedentary behavior and cardiovascular risk. However, to date, no studies have specifically examined sedentary behavior and its relation to the type of fat deposition (e.g., VAT, IMAT). This critical gap in knowledge was recently addressed by a team of researchers who tested the hypothesis that sedentary time would be positively associated with abdominal fat depots, whereas physical activity level would be inversely associated.³⁶ The investigators also predicted that sedentary time and physical activity would be independent, and additive, in their prediction of abdominal fat deposition. The investigators were also interested in potential sex and racial differences between these relations.

Sedentary behavior and physical activity were assessed using self-report questionnaires, and abdominal fat depots were measured using CT scans. The findings provide evidence that sedentary behavior (particularly time spent watching television) and physical activity have distinct, independent associations with patterns of abdominal adipose tissue deposition. Specifically, there appears to be a positive relation between time spent watching television and VAT, independent of total adiposity. The strongest associations were

observed in white men. These findings add to a growing body of literature examining the relations between sedentary behavior, physical activity, and obesity. Future longitudinal studies are necessary to track these associations over time.

Changes in Body Composition with Exercise Training

Regular exercise can alter body composition. Despite the belief that even vigorous exercise burns too few calories to lead to substantial body fat reductions, research has conclusively demonstrated the effectiveness of exercise training in promoting moderate alterations in body composition.

While it is relatively easy to calculate the number of calories a person will expend in an endurance training program and then estimate the total weight loss over a period of time, this type of calculation is rarely accurate because of the interaction of the physiology of energy balance discussed earlier. Changing one (or more) of the three components of energy balance affects the others. Increased energy expenditure in an exercise program that burns about 1,300 kcal per week calculates to a weight loss of approximately 0.15 kg (0.33 lb) per week. However, the simple act of increasing energy expenditure coupled with even small amounts of decreased food intake (or even weight loss) positively affects energy balance. So, simple calculations like the preceding one are rarely accurate, and over long periods of time, individual variation in weight loss from an exercise program can be considerable.

In addition, when significant body weight is lost, the caloric cost of activity (TEA) is lowered proportionately along with RMR. Physiology, appetite, neurohormonal stimuli for food consumption, and even the availability of food all affect the amount of weight loss for any given negative caloric balance. It is simply too difficult and inaccurate to try to calculate these kinds of outcomes over a short-term weight management program. However, most clients in weight management programs want and expect some kind of estimate. Providing that estimate with great caution is well advised.

The ACSM's 2011 position stand on weight management lists several important recommendations for weight loss and weight management.¹¹ One of these is that physical activity is an important

adjunct to a weight management program. The ACSM suggests that 150 to 250 min per week of moderate-intensity physical activity is important to prevent weight gain. However, that volume of activity provides “only moderate weight loss,” and greater volumes of activity are necessary for more significant weight loss. In fact, even more (>250 min per week) may be necessary to prevent weight regain after significant weight loss.

Additionally, examining the energy expended only during exercise does not give the full picture. Metabolic rate remains temporarily elevated after exercise ends. This phenomenon, as mentioned in [chapter 5](#), is referred to as the excess postexercise oxygen consumption (EPOC). Returning the metabolic rate back to its preexercise level can require several minutes following light exercise, such as walking; several hours following very heavy exercise, such as playing a football game; and up to 12 to 24 h or even longer for prolonged, exhaustive exercise, such as running a marathon or ultra-endurance race.

The EPOC can require substantial energy expenditure when considered over the entire recovery period. If, for example, the oxygen consumption following exercise remains elevated by an average of only 0.05 L/min, this will amount to approximately 0.25 kcal/min, or 15 kcal/h. If the metabolism remained elevated for 5 h, this would provide an additional expenditure of 75 kcal that would not normally be included in the calculated total energy expenditure for that particular activity. This additional energy expenditure is ignored in most calculations of the energy costs of various activities. The person in this example, if exercising 5 days per week, would expend 375 kcal, or lose the equivalent of about 0.05 kg (0.1 lb) of fat in 1 week, or 0.45 kg (1.0 lb) in 10 weeks, from the additional caloric expenditure during the recovery period alone.

Studies have shown relatively small but significant changes in both weight and body composition with both aerobic and resistance training, which include

- total weight decrease,
- fat mass and relative body fat decrease, and
- either maintained or increased fat-free mass.

From a summary of hundreds of published studies that monitored body composition changes with aerobic training, the expected changes from a typical 1-year exercise training program (three times per week, 30 to 45 min per day, at 55% to 75% of $\dot{V}O_{2\max}$) would be as follows: -3.2 kg (-7.1 lb) total body mass, -5.2 kg (-11.5 lb) fat mass, and +2.0 kg (+4.4 lb) fat-free mass.³⁷ Relative body fat would decrease by nearly 6% (e.g., from 30% body fat to 24% body fat).

Since the 1990s, abdominal visceral fat ([figure 22.7](#)) has been identified as a major independent risk factor for cardiovascular diseases and obesity. There is now substantial evidence that physical activity reduces the rate of accumulation of visceral fat and that exercise training actually reduces visceral fat stores.³⁴ This could be one of the most important health benefits of an active lifestyle.

Mechanisms for Change in Body Weight and Composition

When attempting to explain how exercise causes such changes in body weight and composition, it is necessary to consider both sides of the energy balance equation. Evaluating energy expenditure requires that we consider each of the three components of energy expenditure: RMR, TEM, and TEA. Evaluating energy intake requires that we also consider the energy that is lost in the feces (energy excreted), which is generally less than 5% of the total caloric intake. Keeping this balance in mind, in the next section we examine some of the possible mechanisms through which exercise might affect body weight and body composition.

Exercise and Appetite

The link between exercise and appetite remains controversial and unresolved. The interrelationships of energy intake, energy expenditure, appetite, and obesity are complex and multifactorial. Issues of hunger, the coupling of energy intake and energy expenditure, the hormones involved with hunger and satiety (the satisfied feeling after a meal), macronutrient intake (dietary pattern), and even differences in responses to exercise between men and women are difficult to sort out. At the present time, we can conclude that (1) exercise, by itself, probably does not significantly stimulate appetite, especially during a weight management program that includes exercise and diet and (2) even if exercise does cause minor

increases in energy intake, this does not preclude its inclusion in weight management programs because the TEA is a major variable that can be altered in these programs.

A classic study by Jean Mayer led him to theorize that energy intake and energy expenditure are related, that is, “coupled within a certain zone” that he called “normal activity.”²⁴ In a study of a large population of factory workers, he and his coauthors found that those who had the highest job-related energy expenditures increased energy intake appropriately to match the energy expenditure. However, the most sedentary workers, who were the heaviest group, had high energy intakes relative to their activity level. Thus, there was a zone of normal activity within which people increased their food intake to accommodate high levels of energy expenditure (work related), but outside that zone (where the relatively sedentary workers fell), energy intake was much higher than necessary by activity standards. Mayer further hypothesized that humans were evolutionarily active (not sedentary); thus, the role of physical activity, perhaps influenced by our genome, functions to increase energy intake only within a small range of caloric expenditure (Mayer’s “normal activity range”).

An additional influential factor in the appetite–weight loss issue is satiety (feeling satisfied after eating), which also affects hunger and energy intake. In a study on exercise, weight loss, and satiety, subjects who lost weight progressively increased their satiety during the training program. Subjects who lost less weight than expected progressively decreased their satiety and concurrently increased their caloric intake during the training program.²⁰ Combining exercise and diet in a weight management program may have some effect on both hunger and satiety, thus affecting how much weight is lost.

The relation between exercise and the hormones associated with appetite and energy intake is part of another complex and as yet unanswered question. Two hormones, leptin and ghrelin (discussed in [chapter 4](#)), have major influences on appetite and satiety, and are therefore significant players in the problem of obesity. These hormones provide acute and chronic feedback to the brain about nutrition status. Ghrelin is produced and secreted by the gastrointestinal system and provides acute input to the brain about

food consumption. Leptin, on the other hand, provides signals related to the long-term energy status associated with energy stores, that is, adipose tissue. Greater fat deposits stimulate greater leptin production. Thus, there is immediate feedback from the gastrointestinal tract about what and how much we eat, plus long-term feedback from the adipocytes about the amount of stored energy. Some people with obesity have been shown to have leptin resistance. Leptin receptors in the central nervous system that were previously sensitive to leptin in the feedback loop become increasingly insensitive to this signaling hormone as adiposity increases.

Exercise and Resting Metabolic Rate

The effects of exercise on the components of energy expenditure became a major topic of interest among researchers in the late 1980s and early 1990s. Of obvious interest is how exercise training might affect the RMR, since RMR represents 60% to 75% of the total calories expended each day. For example, if a 25-year-old man's total daily caloric intake were 2,700 kcal and his RMR accounted for just 60% of that total ($0.60 \times 2,700 = 1,620$ kcal RMR), a mere 1% increase in his RMR would require an extra 16 kcal expenditure each day, or 5,840 kcal per year. This small increase in RMR alone would account for the equivalent of a 0.8 kg (1.7 lb) fat loss per year.

The role of exercise training in increasing RMR has not been totally resolved. Several cross-sectional studies have shown that highly trained athletes have higher RMRs than untrained people of similar age and size. But other studies have not been able to confirm this.³¹ Few longitudinal studies have been conducted to determine the change in RMR in untrained people who undergo training for a period of time, with some suggesting that RMR might increase following training. However, in a study of 40 men and women 17 to 62 years of age (HERITAGE Family Study), a 20-week aerobic training program (three times per week, 35-55 min per day, at 55%-75% of $\dot{V}O_{2\max}$) failed to increase RMR even though $\dot{V}O_{2\max}$ increased by nearly 18%.³⁸ Because RMR is closely related to the fat-free mass of the body (fat-free tissue is more metabolically active), interest has

increased in the use of resistance training to increase fat-free mass in an attempt to increase RMR.

Spot Reduction

Many people, including athletes, believe that exercising a specific area of the body will use the fat in that area, reducing the locally stored fat. Results of several early research studies tended to support this concept of **spot reduction**. But later research suggests that spot reduction is a myth and that exercise, even when localized, draws from almost all of the fat stores of the body, not just from local depots.

One such study used outstanding tennis players, theorizing that they would be ideal subjects for studying spot reduction because they could act as their own controls: Their dominant arms exercise vigorously for several hours every day, whereas their nondominant arms are relatively inactive.¹⁴ The players' dominant arms had substantially greater girths attributable to exercise-induced muscle hypertrophy. But the subcutaneous skinfold fat thickness in the active and inactive arms showed absolutely no differences.

Another study examined the localized effects of a 27-day intense sit-up training program on subcutaneous fat in the abdomen. Researchers found no difference in the rate at which fat cell diameter changed in the abdomen, the subscapular region, and the gluteal region, which indicates a lack of adaptation at the training site (abdomen).¹⁹ Decreases in girth can occur with exercise training, but these primarily result from increased muscle tone, not fat loss.

Low-Intensity Aerobic Exercise

As we discussed in earlier chapters, the higher the exercise intensity, the greater the body's reliance on carbohydrate as an energy source. With high-intensity aerobic exercise, carbohydrate can supply up to 90% or more of the body's energy needs. During the late 1980s, various groups promoted **low-intensity aerobic exercise** to increase the loss of body fat. These groups theorized that low-intensity aerobic training would allow the body to use more fat as the energy source, hastening the loss of body fat. Indeed, the body uses a higher percentage of fat for energy at lower exercise intensities. However, the total calories expended does not necessarily change as a result of the body's use of fat.

This is illustrated in [table 22.4](#). In this hypothetical example, a 23-year-old woman with a $\dot{V}O_{2\max}$ of 3.0 L/min exercises for 30 min at 50% of her $\dot{V}O_{2\max}$ on one day and for 30 min at 75% of her $\dot{V}O_{2\max}$ on another. The total calories from fat do not differ between the low- and high-intensity aerobic workouts: In both cases, she burns about 110 kcal of fat during 30 min. Most importantly, however, if she is fit enough to sustain the higher-intensity workout, she will expend about 50% more total calories for the same time period.

Scientists have determined that there is an optimal zone within which rates of fat oxidation are at their highest. The $\text{Fat}_{\max}$ zone, defined as that zone where fat oxidation rates are within 10% of the peak rate, was found to vary from between 55% and 72% of $\dot{V}O_{2\max}$.¹ This is illustrated in [figure 22.8](#). However, as already noted, this high rate of fat oxidation does not necessarily mean a higher total calorie expenditure or greater contribution to weight loss.

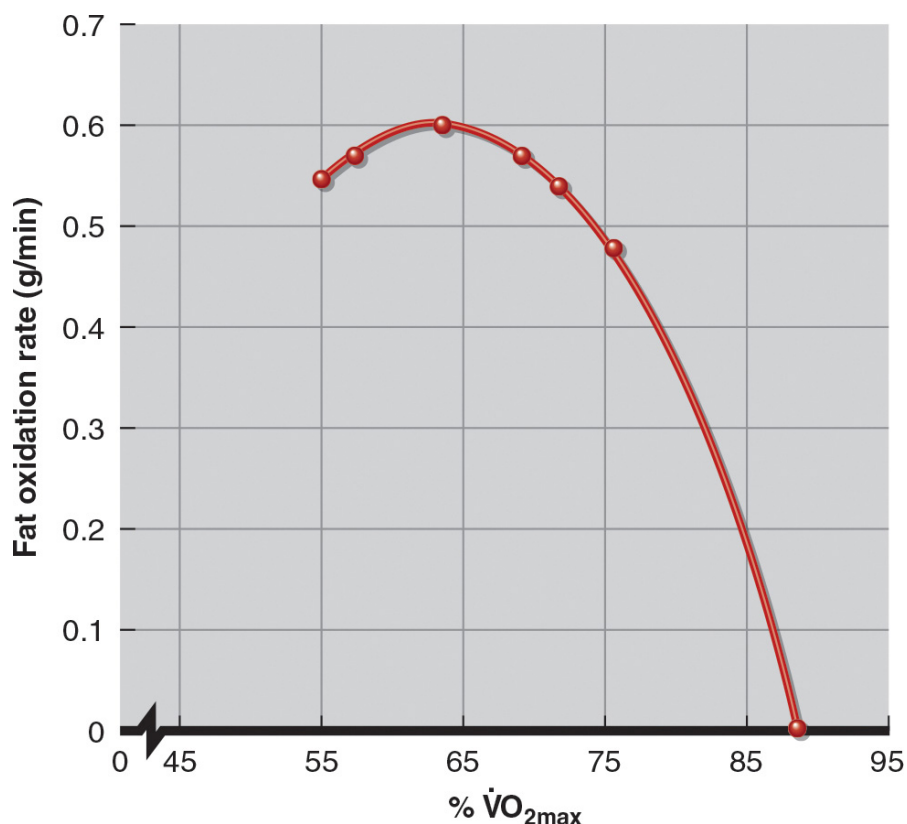


FIGURE 22.8 Rate of fat oxidation at various exercise intensities, expressed as a percentage of $\dot{V}O_{2\max}$.

Reprinted by permission from J. Acton, M. Gleeson, and A.E. Jeukendrup, "Determination of the Exercise Intensity That Elicits Maximal Fat Oxidation," *Medicine and Science in Sports and Exercise* 34 (2002): 92-97.

TABLE 22.4 Estimation of Kilocalories Used From Fat and Carbohydrate for a Low- and High-Intensity Aerobic Training Bout of 30 Minutes Duration

Exercise intensity	Average $\dot{V}O_2$ (L/min)	Average RER	% kcal CHO	% kcal fat	kcal for 30 min CHO	kcal for 30 min fat	kcal for 30 min total
Low, 50%	1.50	0.85	50	50	110	110	220
High, 75%	2.25	0.90	67	33	222	110	332

Note. RER = respiratory exchange ratio; CHO = carbohydrate. Subject was a fit but not highly trained 23-year-old woman ($\dot{V}O_{2\max} = 3.0$ L/min).

In Review

- Inactivity is a major cause of obesity and type 2 diabetes in the United States; it is perhaps as significant as overeating.
- A minimum of 150 to 250 min per week of moderate-intensity physical activity is important to prevent weight gain. Greater volumes of activity are necessary for significant weight loss.
- The energy expended by activity includes the rate of energy expenditure during the activity and also the energy expended after the exercise, because the metabolic rate remains elevated for some time after the activity ends. This phenomenon is known as excess postexercise oxygen consumption, or EPOC.
- Dietary restriction alone may cause fat loss, but fat-free mass is also lost. With exercise, either alone or with diet, fat is lost, but fat-free mass is either maintained or increased. Possibly the greatest benefit of physical activity and formal exercise is their role in attenuating the accumulation of visceral fat or in reducing visceral fat stores.
- Simply being active, independent of a formal exercise program, is important in the prevention of obesity.
- Energy intake – energy expended = RMR + TEM + TEA when one is in energy balance.
- A certain amount of activity, Mayer's "normal zone," appears to be necessary for the body to balance energy intake and expenditure.
- Research indicates that exercise may suppress appetite, but this issue is controversial.
- A single bout of exercise increases the TEM.
- Exercise increases lipid mobilization from adipose tissue.
- Spot reduction is a myth.
- Low-intensity aerobic exercise burns no more fat than more vigorous exercise, and more total calories are burned in a more strenuous workout.

Understanding Diabetes

Diabetes mellitus, also called diabetes, is a disease characterized by high blood glucose concentrations (hyperglycemia) resulting from either an inadequate production of insulin by the pancreas, an inability of insulin to facilitate the transport of glucose into the cells, or both. Recall from [chapter 4](#) that insulin is a hormone that reduces circulating glucose by facilitating its transport into the cells. We first address the terminology used in defining diabetes and disordered blood glucose control (glycemic control) and then look at the prevalence of diabetes in the United States.

Terminology and Classification

Diabetes mellitus is really two separate diseases—type 1 diabetes mellitus (previously known as juvenile-onset diabetes, or insulin-dependent diabetes mellitus) and type 2 diabetes mellitus (previously adult-onset diabetes).²² Type 2 diabetes is closely associated with obesity, and as is the case with obesity, there has been a near epidemic of type 2 diabetes in the United States in children and adults.

Type 1 diabetes is caused by the failure of the **β -cells** in the pancreas to produce sufficient insulin. This condition is the result of the body's immune system destroying pancreatic β -cells (a process called apoptosis). As they decrease in number, less insulin is produced, and at some critical tipping point, there is insufficient insulin for proper control of blood glucose. Thus, this type has also been referred to as **insulin-dependent diabetes mellitus (IDDM)**. Type 1 diabetes accounts for only 5% to 10% of all cases of diabetes.

Type 2 diabetes is the result of the failure of insulin to facilitate the transport of glucose into the cells and is a result of insulin resistance. Type 2 diabetes accounts for 90% to 95% of all cases of diabetes. Insulin's primary function is to facilitate the transport of glucose from the blood across the cell membrane. *Insulin resistance* refers to the condition in which a normal insulin concentration in the blood produces a less than normal biological response. The body needs more insulin to transport a given amount of glucose into the cell. *Insulin sensitivity* is a related concept; it provides an index of the

effectiveness of a given insulin concentration in the blood. Insulin resistance increases as type 2 diabetes develops and progresses.

A third type of diabetes, gestational diabetes, is a form of diabetes that develops in women and their fetuses during pregnancy and occurs in about 4% of all pregnancies. Fortunately, it usually disappears in both mother and baby after delivery. Unfortunately, gestational diabetes can lead to complications during pregnancy.

Prediabetes refers to the condition in those who have impaired fasting glucose, impaired glucose tolerance, or both. Both type 1 and type 2 diabetes are diagnosed on the basis of a plasma glucose level greater than 125 mg/dl following an 8 h fast. **Impaired fasting glucose** is defined as a plasma glucose level between 100 and 125 mg/dl, again following an 8 h fast. **Impaired glucose tolerance** is determined by a glucose tolerance test. This involves drinking a solution in which 75 g of anhydrous glucose is dissolved in water. Plasma glucose concentrations are measured 2 h later. A glucose concentration of 200 mg/dl or higher is diagnostic of diabetes. Values of 140 to 199 mg/dl represent impaired glucose tolerance, and values below 140 mg/dl are considered normal.

For research studies, an intravenous glucose tolerance test (IVGTT) can also be used. A catheter is placed in each arm, with the glucose solution injected into a vein in one arm, and blood samples are withdrawn from a vein in the other arm over the course of 3 h. Samples are taken more frequently during the first 15 to 45 min of the test and less frequently from 60 to 180 min. This allows one to develop a curve of both the glucose and insulin responses to the injected glucose load. The IVGTT is more precise than the oral glucose tolerance test.

There are common symptoms that can be used to identify those at risk for diabetes. They include the following:

Type 1 Diabetes

- Frequent urination
- Excessive or unusual thirst
- Unusual, unexplained weight loss
- Extreme hunger
- Extreme fatigue and irritability

Type 2 Diabetes

- Any of the type 1 symptoms
- Frequent infections
- Blurred vision or sudden vision changes
- Tingling or numbness in hands or feet
- Cuts, bruises, or sores that are slow to heal
- Recurring skin, gum, or bladder infections

RESEARCH PERSPECTIVE 22.3

Metformin or Exercise or Both to Treat Diabetes?

Type 2 diabetes and prediabetes are risk factors for the development of cardiovascular disease. Although the causes of type 2 diabetes are multifactorial, insulin resistance (and the resulting hyperinsulinemia) is likely a primary contributor. Thus, targeting insulin resistance is important for the management and prevention of type 2 diabetes. Diet and exercise are the first-line therapy for improving glycemic control; however, long-term adherence to exercise programs remains poor, and many individuals also require pharmacological therapy to maintain normal blood sugar concentrations.

The American Diabetes Association recently recommended that, in addition to lifestyle modifications, individuals with prediabetes should be considered for metformin therapy (the most widely used medication for diabetes treatment, which decreases liver glucose production). The underlying assumption behind this recommendation is that because metformin and exercise each lower blood glucose, the actions of metformin and the benefits of physical activity will be *additive*, such that combined treatment will be more beneficial than either treatment alone. However, because the generation of reactive oxygen species (ROS) is a primary factor promoting the metabolic adaptation to exercise and because metformin reduces ROS generation, metformin may actually *attenuate* metabolic adaptation to exercise via an oxidative stress mechanism. So, does combining metformin with exercise have additive, opposing, or no benefit for glucose regulation and cardiovascular disease risk reduction?

The answer to this seeming paradox was recently reviewed by Malin and Braun.²³ Work from a series of studies published by their laboratory suggests that combining the two metabolically beneficial treatments does not necessarily result in an additive outcome. They conducted a prospective, double-blind, randomized controlled trial to examine the long-term effects of combining exercise training and metformin on insulin sensitivity in men and women with prediabetes. Participants were assigned to four different treatment groups: placebo, metformin, exercise training plus placebo, or

exercise training plus metformin. All treatments reduced fasting plasma insulin. Metformin alone or exercise training alone increased skeletal muscle insulin sensitivity. However, metformin blunted the effect of exercise training on skeletal muscle insulin sensitivity by ~30%, suggesting that metformin dampens some of the beneficial effects of exercise training in regards to glucose regulation.

This leaves clinicians with the question of whether they should coprescribe metformin with exercise in order to treat patients or whether exercise treatment alone is sufficient to lower diabetes risk. Not surprisingly, the answer is complicated. Research studies have suggested that metformin blunts the effects of exercise on improving insulin sensitivity, metabolic syndrome prevalence, and inflammation. Although these effects may seem concerning, it is important to realize that metformin has distinct effects on multiple tissues. Thus, the degree of additive benefit will depend, in large extent, on the tissue of interest. Nevertheless, the authors argue that exercise alone is the best first-line therapy for reducing the risk of type 2 diabetes.

Prevalence of Diabetes

According to the American Diabetes Association, in 2015, approximately 30 million Americans had diabetes and an estimated 7 million people were undiagnosed. Additionally, an estimated 84 million adults had prediabetes. The prevalence of diabetes increased from 4.9% of the U.S. population in 1990 to 9.4% in 2015. Approximately 25% of people over 65 have diabetes, and diabetes remains the seventh leading cause of death in the United States. Further, the prevalence of diagnosed diabetes in those aged 20 years or older was 7.4% for non-Hispanic whites, 8.0% for Asian Americans, 12.7% for non-Hispanic blacks, and 12.1% for Hispanics and Latinos.² These ethnic differences in diabetes rates parallel the differences in obesity rates discussed earlier in this chapter.

The true prevalence of type 2 diabetes in children is still low, but prevalence has been estimated to have increased by as much as 10-fold over the last 20 years. Furthermore, in 10- to 19-year-olds, studies have shown that type 2 diabetes accounted for between 33% and 46% of all diabetes.⁴¹ This is very disturbing, considering that not too long ago we referred to type 2 diabetes as adult-onset diabetes.

Etiology of Diabetes

Hereditry appears to play a major role in both type 1 and type 2 diabetes. With type 1 diabetes, the β -cells (insulin-secreting cells) of

the pancreas are destroyed. This destruction may be caused by the body's immune system, increased β -cell susceptibility to viruses, or β -cell degeneration. Type 1 diabetes typically has a sudden onset during childhood or young adulthood. This leads to almost total insulin deficiency, and daily injections of insulin are usually required to control the disease.

In type 2 diabetes, the onset of the disease is more gradual, and the causes are more difficult to establish. Type 2 diabetes is usually characterized by three major metabolic abnormalities: insulin resistance (impaired insulin action), primarily in skeletal muscle tissue, increased insulin secretion (hyperinsulinemia in response to the increased insulin resistance), or excessive glucose output from the liver, or some combination of these.

Obesity plays a major role in the development of type 2 diabetes. With obesity, insulin resistance is commonly a primary initial problem. The β -cells of the pancreas respond to the increased blood glucose (secondary to the insulin resistance) by producing and secreting more insulin. Thus, it is not uncommon for people with recently diagnosed type 2 diabetes to have both hyperglycemia (high blood glucose) and hyperinsulinemia (high blood insulin). This is all a result of insulin receptors throughout the body (especially in skeletal muscle) that are less effective in facilitating the transport of glucose into the cell.

Health Problems Linked to Diabetes

Considerable health risks are associated with diabetes. People with this disease have a relatively high mortality rate. Diabetes places a person at increased risk for

- coronary artery or heart disease and peripheral vascular disease;
- hypertension, cerebrovascular disease, and stroke;
- kidney disease;
- nervous system disease;
- eye disorders, including blindness;
- dental disease;
- amputations; and
- complications during pregnancy.

Scientists have established the interrelatedness among coronary artery disease, hypertension, obesity, and type 2 diabetes. **Hyperinsulinemia** (high blood levels of insulin) and insulin resistance appear to be the important threads linking these disorders, possibly through insulin-mediated sympathetic nervous system stimulation (increased insulin levels cause increased sympathetic nervous system activity), inflammation, metabolic dysfunction, or some combination of these. Again, obesity seems to be the trigger setting off this reaction.

Treatment of Diabetes

The major modes of treatment for type 1 diabetes are insulin administration, diet, and exercise. The dosage of insulin is adjusted to allow normal carbohydrate, fat, and protein metabolism and normal levels of blood glucose. The type of insulin injected—short acting or intermediate acting—and the time of day at which the injections are administered are also individualized to maintain glycemic control throughout the day.

For type 2 diabetes, the focus has traditionally been on three factors: weight loss, a healthy dietary pattern, and exercise and physical activity. Treatment of diabetes (especially type 2 diabetes) with drugs has expanded significantly over the past 25 years. Drugs that stimulate production and release of insulin, inhibit production and release of glucose from the liver, or block enzymes in the stomach that break down carbohydrates and increase insulin sensitivity are in common use.

A balanced, healthy diet generally is prescribed for people with diabetes. In the past, a low-carbohydrate diet was recommended to better control blood sugar levels. However, a low-carbohydrate diet is usually associated with an increase in dietary fat, which can have a major negative effect on blood lipid levels. Because people with diabetes often have abnormal lipids and are already at greater risk for coronary artery disease, this is not desirable. Maintenance of proper blood sugar levels may be difficult in patients with obesity, so a reduced-calorie diet is used in order to facilitate significant loss of body fat. For many people with type 2 diabetes, weight loss alone can bring blood sugar levels back into the normal range. Additionally,

since regular exercise and physical activity can increase insulin sensitivity, increased physical activity and a healthy dietary pattern are both very important aspects of the treatment plan for the person with type 2 diabetes.

During the mid-1990s, the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) of the National Institutes of Health designed and initiated a study to determine whether either a lifestyle intervention (diet and exercise) or an oral diabetes drug (metformin) could prevent or delay the onset of type 2 diabetes in people with impaired glucose tolerance (prediabetes). A total of 3,234 persons 25 years of age and older were randomly assigned to placebo, drug, or lifestyle groups. The goals for the lifestyle modification program were to lose at least 7% of body weight and to participate in physical activity for at least 150 min per week. Both interventions proved very successful, delaying the development of diabetes by 11 years in the lifestyle group and by 3 years in the metformin group.²¹ This landmark study clearly illustrates the power of lifestyle intervention in reducing the risk of a major debilitating disease and its serious consequences. The Finnish Diabetes Prevention Study confirmed those results and further reinforced the value of physical activity for prevention of diabetes.³⁵

Role of Physical Activity in Diabetes

Indirect scientific evidence has clearly established that a physically active lifestyle reduces the risk of developing type 2 diabetes,³ but there is little evidence to support this for type 1 diabetes. However, most physicians and scientists agree that physical activity is an important part of the treatment plan for both types of diabetes. Because there is such a disparity between the characteristics and responses of those with type 1 and type 2 diabetes, we discuss them separately.



Type 1 Diabetes

The role of regular exercise and physical training in improving glycemic control (regulation of blood sugar levels) in patients with type 1 diabetes has not been clearly defined and is controversial. The most distinguishing feature differentiating type 1 and type 2 diabetes is that those with type 1 have low blood insulin levels attributable to the inability or reduced ability of the pancreas to produce insulin. Those with type 1 diabetes are prone to hypoglycemia (low blood sugar levels) during and immediately after exercise because the liver fails to release glucose at a rate that can keep up with glucose utilization. For these people, exercise can lead to excessive swings in plasma glucose levels that are unacceptable for the management of the disease. The degree of glycemic control during exercise varies tremendously among individuals with type 1 diabetes. As a result, exercise and exercise training can improve glycemic control in some patients, mainly those who are less prone to hypoglycemia, but not in others.

Although glycemic control is generally not improved by exercise in most people with type 1 diabetes, there are other potential benefits of exercise for these patients. These patients have a risk for coronary artery disease two to three times greater, and exercise may help reduce this risk. Exercise may also help reduce the risk for cerebrovascular and peripheral arterial diseases.

People with uncomplicated type 1 diabetes do not have to restrict physical activity provided that blood sugar levels are controlled

appropriately. A number of athletes who have type 1 diabetes have trained and competed successfully. Monitoring blood sugar levels in an exercising person with type 1 diabetes is important so that diet and insulin dosages can be altered accordingly. This is particularly important for those who are competing at high intensities or for extended periods of time.

Special attention also should be given to the feet of people with diabetes, since it is common for them to experience peripheral neuropathy (diseased nerves) with some loss of sensation in the feet. Peripheral vascular disease is also more common in patients with diabetes, so the circulation to the extremities, especially the feet, often is impaired significantly. Ulcerations and other lesions on the feet account for more than half of all hospitalizations of diabetic patients. Because weight-bearing exercise places additional stress on the feet, the proper selection of footwear and appropriate preventive foot care are important.

Type 2 Diabetes

Exercise plays a major role in glycemic control for people with type 2 diabetes. Insulin production is generally not of concern in this group, particularly during the early stages of the disease, so the major problem with this form of diabetes is the lack of target cell response to insulin (insulin resistance). Because the cells become resistant to insulin, the hormone cannot perform its function of facilitating glucose transport across the cell membrane. The act of muscle contraction has an insulin-like effect. Membrane permeability to glucose increases with muscular contraction, likely attributable to an increase in the number of GLUT-4 glucose transporters associated with the plasma membrane.¹⁷ Thus, acute bouts of exercise decrease insulin resistance and increase insulin sensitivity. This reduces the cells' requirements for insulin. Resistance and aerobic training appear to produce similar effects,⁴⁰ although some evidence suggests that combining resistance and aerobic exercise is the optimal exercise strategy for reducing insulin resistance.¹⁰ This decrease in insulin resistance and increase in insulin sensitivity may primarily be a response to each individual bout of exercise rather than the result of a long-term change associated with training. Some studies have shown that the effect dissipates within 72 h.

The American Diabetes Association recently released an updated position statement on physical activity and exercise and diabetes.⁸ This position statement provides evidence-based recommendations regarding physical activity and exercise for people with type 1 diabetes and type 2 diabetes. Recommendations from the ADA are summarized in [table 22.5](#).

In Review

- Type 1 diabetes involves destruction of the β -cells in the pancreas, and it typically has an early onset, that is, in adolescence or at younger ages. Type 2 diabetes is characterized by impaired glucose tolerance because of insulin resistance.
- Diabetes is characterized by hyperglycemia. It develops from inadequate insulin utilization (insulin resistance), secretion, or both.
- Major modes of treatment for diabetes are drugs (if needed), diet, and, especially for type 2 diabetes, exercise and weight loss.
- In people with type 1 diabetes, glycemic control may or may not be improved with exercise. They have a greater risk for coronary artery disease, and exercise can decrease that risk.
- People with type 1 diabetes must carefully monitor blood glucose before, during, and after exercise so that diet and insulin dosage can be altered as needed.
- Peripheral neuropathy is a serious complication associated with type 1 and longstanding type 2 diabetes. Those with peripheral neuropathy should give their feet special attention because of the loss of sensation and impaired peripheral circulation that accompany it.
- Type 2 diabetes responds well to exercise. Glucose utilization and insulin sensitivity improve with exercise, most likely because of an increase in GLUT-4 receptors in skeletal muscle.

TABLE 22.5 Summary of Recommendations by the American Diabetes Association for Reducing Sedentary Time and Increasing Physical Activity in People with Diabetes

Recommendations for reduced sedentary time
All adults, and particularly those with type 2 diabetes, should decrease the amount of time spent in daily sedentary behavior. Prolonged sitting should be interrupted with bouts of light activity every 30 min for blood glucose benefits, at least in adults with type 2 diabetes. The above two recommendations are additional to, and not a replacement for, increased structured exercise and incidental movement.
Physical activity and type 2 diabetes
Daily exercise, or at least not allowing more than 2 days to elapse between exercise sessions, is recommended to enhance insulin

action.

Adults with type 2 diabetes should ideally perform both aerobic and resistance exercise training for optimal glycemic and health outcomes.

Children and adolescents with type 2 diabetes should be encouraged to meet the same physical activity goals set for youth in general.

Structured lifestyle interventions that include at least 150 min/week of physical activity and dietary changes resulting in weight loss of 5% to 7% are recommended to prevent or delay the onset of type 2 diabetes in populations at high risk and with prediabetes.

Physical activity and type 1 diabetes

Youth and adults with type 1 diabetes can benefit from being physically active, and activity should be recommended to all. Blood glucose responses to physical activity in all people with type 1 diabetes are highly variable based on activity type/timing and require different adjustments.

Additional carbohydrate intake or insulin reductions are typically required to maintain glycemic balance during and after physical activity. Frequent blood glucose checks are required to implement strategies for adjusting carbohydrate intake and insulin dose. Insulin users can exercise using either basal-bolus injection regimens or insulin pumps, but there are advantages and disadvantages to both insulin delivery methods. Continuous glucose monitoring during physical activity can be used to detect hypoglycemia when used as an adjunct rather than in place of capillary glucose tests.

Recommended physical activity participation for people with diabetes

Preexercise medical clearance is generally unnecessary for asymptomatic individuals prior to beginning low- or moderate-intensity physical activity not exceeding the demands of brisk walking or everyday living.

Most adults with diabetes should engage in 150 min or more of moderate- to vigorous-intensity activity weekly, spread over at least 3 days/week, with no more than 2 consecutive days without activity. Shorter durations (minimum 75 min/week) of vigorous-intensity activity or interval training may be sufficient for younger and more physically fit individuals.

Children and adolescents with type 1 or type 2 diabetes should engage in 60 min/day or more of moderate- or vigorous-intensity aerobic activity, with vigorous, muscle-strengthening, and bone-strengthening activities included at least 3 days/week.

IN CLOSING

In these last two chapters, we have concluded our look at the role of physical activity in preventing and treating coronary artery disease, hypertension, obesity, and diabetes. We have seen that exercise can decrease individual risk and also can be an integral part of treatment, improving overall health as well as alleviating some symptoms.

With this chapter, we also conclude our journey to an understanding of exercise and sport physiology. We began this book by reviewing how various body systems function during exercise and how they respond to chronic training. We saw how physical activity and performance are affected by the environment, such as extremes of heat, cold, and barometric pressure. We turned our attention to ways in which athletes can, or can attempt to, optimize their performance. Then we considered the unique differences for older and younger participants and male and female participants in sport and exercise. And finally, we examined the role of exercise in the maintenance of health and the development of fitness.

It has been a long journey from cover to cover, but we hope that you will close this book with a new appreciation of physical activity. Perhaps you will leave this book with a new awareness of how your body performs physical activity. Maybe, if you have not yet done so, you will feel compelled to commit to a personal exercise program. And we hope that you now feel some excitement about exercise and sport physiology, realizing that these areas of study affect so many aspects of our lives.

KEY TERMS

β -cells

body mass index (BMI)

diabetes mellitus

hyperinsulinemia

impaired fasting glucose

impaired glucose tolerance

insulin-dependent diabetes mellitus (IDDM)

lower body (gynoid) obesity

low-intensity aerobic exercise

metabolic syndrome

obesity

overweight

prediabetes

spot reduction

thermic effect of activity (TEA)

thermic effect of a meal (TEM)

type 1 diabetes
type 2 diabetes
upper body (android) obesity

STUDY QUESTIONS

1. What is the difference between overweight and obesity?
2. What are the methods for determining body composition, and what are the strengths and weakness of each?
3. What is body mass index? What is its significance?
4. What is the prevalence of obesity in the United States today? Is there a difference in the prevalence between men and women? Children and adults? Blacks and whites?
5. What are several health-related problems associated with obesity?
6. How are obesity, coronary artery disease, hypertension, insulin resistance, and diabetes interrelated?
7. Describe several methods for treating obesity. Which are the most effective?
8. What role does exercise play in the prevention and treatment of obesity?
9. By what mechanisms might exercise effect losses in total weight and fat weight?
10. How effective is spot reduction? Low-intensity aerobic exercise?
11. Describe the two major types of diabetes. How are they caused?
12. What health risks are associated with diabetes?
13. Discuss the major points published in the *Summary of Recommendations by the American Diabetes Association for Reducing Sedentary Time and Increasing Physical Activity in Diabetes* as they relate to exercise.
14. Describe the role of exercise in preventing diabetes and in treating patients with type 1 and type 2 diabetes.

STUDY GUIDE ACTIVITIES

In addition to the activities listed in the chapter opening outline, two other activities are available in the web study guide, located at

www.HumanKinetics.com/PhysiologyOfSportAndExercise

The **KEY TERMS** activity reviews important terms, and the end-of-chapter **QUIZ** tests your understanding of the material covered in the chapter.